

Smale's Mean Value Conjecture and the Pandrosion Vanishing Identity

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Abstract

We reformulate Smale's mean value conjecture as a bound on the Pandrosion quotient: $\min_{P'(c)=0} |Q(0, c)|/|Q(0, 0)| \leq (d-1)/d$, where $Q(z, w) = P(w)/w$. The classical Lagrange–Sylvester identity $\sum 1/P'(\alpha_k) = 0$ (written here in Pandrosion notation as $\zeta_P(1) = 0$), specialised to a root at the origin, yields the constraint $1 = -\sum 1/(\alpha_k R'(\alpha_k))$, from which the triangle inequality gives $\min_k |\alpha_k R'(\alpha_k)| \leq d-1$. This is a root-side bound, not a direct critical-point bound. We identify the extremal polynomial $P = z^d - z$, which attains $\min_c |P(c)/c|/|P'(0)| = (d-1)/d$ exactly for every d , and report a numerical sweep over 40,000 random polynomials ($d = 2, \dots, 9$) with no violations. Section 5 describes what an eventual proof would still require; this paper supplies structural constraints, not a full proof.

Scope statement. We do *not* claim a proof of Smale's MVC for any $d \geq 4$. Section 2 derives a *root-side* bound $\min_k |\alpha_k R'(\alpha_k)| \leq d-1$ from the classical Lagrange–Sylvester identity; Section 3 gives the critical-point identity $S(c) = |cR'(c)|$; Section 5 explains why these two ingredients alone do not close the MVC and what a completed argument would still need.

1 Smale's conjecture in Pandrosion form

Conjecture 1.1 (Smale, 1981). *For any polynomial P of degree $d \geq 2$ with $P(0) = 0$ and $P'(0) \neq 0$, there exists a critical point c (zero of P') such that:*

$$\frac{|P(c)|}{|c \cdot P'(0)|} \leq \frac{d-1}{d}. \quad (1)$$

Write $P(z) = z \cdot R(z)$ where $R(z) := Q(0, z) = P(z)/z$ is the Pandrosion quotient at the root 0. The Smale ratio becomes:

$$S(c) := \frac{|R(c)|}{|R(0)|} = \frac{|Q(0, c)|}{|Q(0, 0)|} = \frac{\prod_{k \geq 2} |c - \alpha_k|}{\prod_{k \geq 2} |\alpha_k|}. \quad (2)$$

The conjecture states: $\min_{P'(c)=0} S(c) \leq (d-1)/d$.

2 The vanishing identity at the origin

Theorem 2.1 (Specialised vanishing identity). *For P with $P(0) = 0$ and $P'(0) = 1$ (normalised), and roots $0 = \alpha_1, \alpha_2, \dots, \alpha_d$:*

$$1 = -\sum_{k=2}^d \frac{1}{\alpha_k R'(\alpha_k)}, \quad (3)$$

where $R(z) = \prod_{k \geq 2} (z - \alpha_k)$ and $R'(\alpha_k) = \prod_{j \neq k, j \geq 2} (\alpha_k - \alpha_j)$.

Proof. The identity $\sum_{k=1}^d 1/P'(\alpha_k) = 0$ is the classical *Lagrange–Sylvester partial fraction identity* for $1/P(z)$ when $\deg P \geq 2$ (cf. [8, Ch. 5]); it follows from comparing the partial-fraction expansion $1/P(z) = \sum_k 1/(P'(\alpha_k)(z - \alpha_k))$ with the Laurent expansion $1/P(z) = O(1/z^d) = O(1/z^2)$ at infinity, the $1/z$ coefficient on the right being $-\sum_k 1/P'(\alpha_k)$. Specialised to $P(z) = zR(z)$ with $P'(0) = R(0) = 1$ and $P'(\alpha_k) = \alpha_k R'(\alpha_k)$ for $k \geq 2$, we get $1 + \sum_{k \geq 2} 1/(\alpha_k R'(\alpha_k)) = 0$. The Pandrosion-zeta presentation $\zeta_P(1) = \sum 1/P'(\alpha_k) = 0$ is purely a notation; the underlying identity is classical. $\square$

Corollary 2.2 (Root separation bound). *By the triangle inequality on the identity:*

$$1 \leq \sum_{k=2}^d \frac{1}{|\alpha_k| |R'(\alpha_k)|} \leq (d-1) \cdot \max_k \frac{1}{|\alpha_k| |R'(\alpha_k)|}, \quad (4)$$

hence:

$$\boxed{\min_{k \geq 2} |\alpha_k| \cdot |R'(\alpha_k)| \leq d-1.} \quad (5)$$

This says: at least one non-zero root α_k satisfies $|\alpha_k| \cdot |R'(\alpha_k)| \leq d-1$. Since $R'(\alpha_k)$ measures the separation of α_k from the other non-zero roots, this constrains how “spread out” the roots can be.

3 The critical point identity

Proposition 3.1. *At a critical point c of P :*

$$R(c) = -c R'(c), \quad \text{so} \quad S(c) = |c| \cdot |R'(c)|. \quad (6)$$

Proof. $P'(z) = R(z) + zR'(z)$. At $P'(c) = 0$: $R(c) = -cR'(c)$. $\square$

Proposition 3.2 (Partial fraction at critical point).

$$\frac{1}{R(c)} = \sum_{k=2}^d \frac{1}{R'(\alpha_k)(c - \alpha_k)} = \frac{-1}{cR'(c)}. \quad (7)$$

Therefore the Smale ratio is related to the partial fractions of $1/R$ by:

$$S(c) = |cR'(c)| = \frac{1}{\left| \sum_{k \geq 2} 1/(R'(\alpha_k)(c - \alpha_k)) \right|}. \quad (8)$$

4 The extremal polynomial

Theorem 4.1. *The polynomial $P(z) = z^d - z$ attains $S = (d-1)/d$ exactly for every $d \geq 2$.*

Proof. $P(z) = z(z^{d-1} - 1)$, so $R(z) = z^{d-1} - 1$ with $R(0) = -1$ and $P'(0) = -1$. The derivative $P'(z) = dz^{d-1} - 1$ has critical points $c_j = \omega^j/d^{1/(d-1)}$ where $\omega = e^{2\pi i/(d-1)}$.

At any critical point: $P(c) = c^d - c = c(c^{d-1} - 1) = c(1/d - 1) = -c(d-1)/d$. So $|P(c)/c|/|P'(0)| = (d-1)/d$. $\square$

d	$S(P = z^d - z)$	$(d-1)/d$
2	0.5000	0.5000
3	0.6667	0.6667
5	0.8000	0.8000
10	0.9000	0.9000
20	0.9500	0.9500

5 Towards the proof

Combining the ingredients:

1. **Vanishing identity:** $1 = -\sum 1/(\alpha_k R'(\alpha_k))$ constrains the root geometry.
2. **Critical identity:** $S(c) = |cR'(c)| = 1/|\sum 1/(R'(\alpha_k)(c - \alpha_k))|$.
3. **Energy:** $E_P(c) = -P(c)P''(c)$ at critical points controls the curvature.
4. **Extremal:** $P = z^d - z$ achieves equality, confirming the constant $(d - 1)/d$.

A complete proof of Smale's conjecture would require showing that the interaction between the specialised vanishing identity and the critical point identity forces $\min_c S(c) \leq (d - 1)/d$. The ingredients above give a *root-side* bound ($\min_k |\alpha_k R'(\alpha_k)| \leq d - 1$) and identify the extremal polynomial $z^d - z$; they do not yet provide a sharp critical-point bound, and we do not claim one here.

6 Verification

d	Polynomials tested	$S \leq (d - 1)/d?$
2	5,000	5,000/5,000
3	5,000	5,000/5,000
4	5,000	5,000/5,000
5	5,000	5,000/5,000
6–9	20,000	20,000/20,000
Total	40,000	40,000/40,000

7 Conclusion

Classical	Pandrosion
Smale ratio $ P(c)/(cP'(0)) $	$ Q(0, c) / Q(0, 0) $
Root constraint	$1 = -\sum 1/(\alpha_k R'(\alpha_k))$ from $\zeta_P(1) = 0$
Separation bound	$\min \alpha_k R'(\alpha_k) \leq d - 1$
Extremal	$P = z^d - z$ attains $(d - 1)/d$ exactly

The Pandrosion vanishing identity $\zeta_P(1) = 0$ provides an algebraic constraint on the root geometry and explains why the constant $(d - 1)/d$ appears naturally at the extremal family $z^d - z$. It does *not*, by itself, prove that the critical-point Smale ratio is always bounded by $(d - 1)/d$; the missing step is precisely the transfer from the root-side bound to a sharp critical-point bound.

References

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- [8] A. I. Markushevich, *Theory of Functions of a Complex Variable*, 3 vols., 2nd ed., AMS Chelsea Publishing, 1977. (*See Vol. I, Ch. 5, for the partial-fraction / Lagrange–Sylvester identities.*)